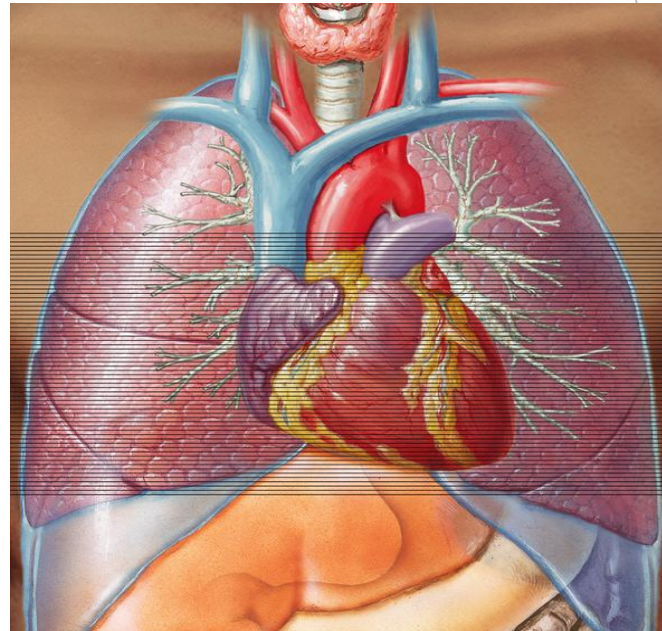


Principles of Cardiothoracic Surgery

Dr/ Montaser El sawy
Professor of Cardiothoracic Surgery-2022
MD in Cardiothoracic Surgery-2010
Master Degree in General Surgery-2005



I-Principles of Thoracic surgery

PRIMARY LUNG CANCER

Aetiology

Predisposing factors:

- Cigarette smoking.
- Air pollution.
- Exposure to uranium, chromium, arsenic, haematite and asbestos.

Pathology

Non-small cell lung cancer (NSCLC) (85%)

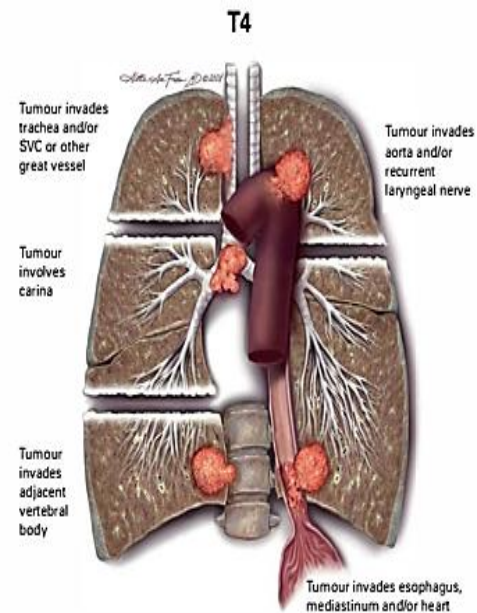
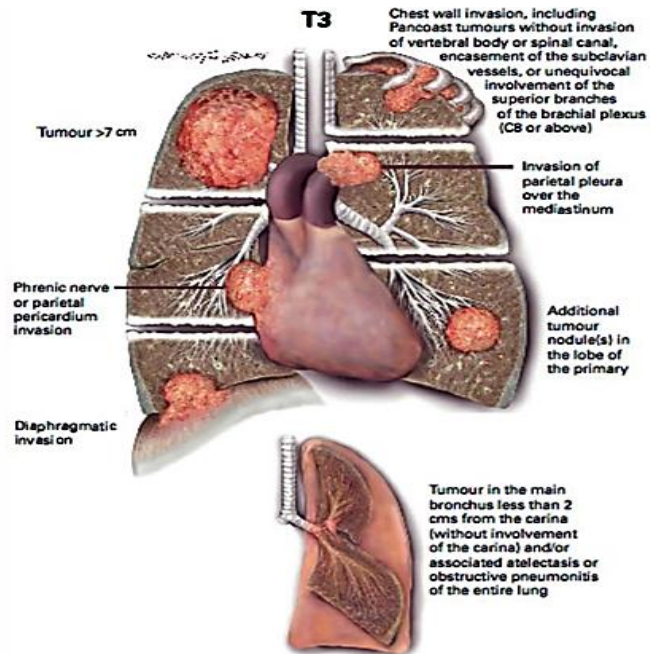
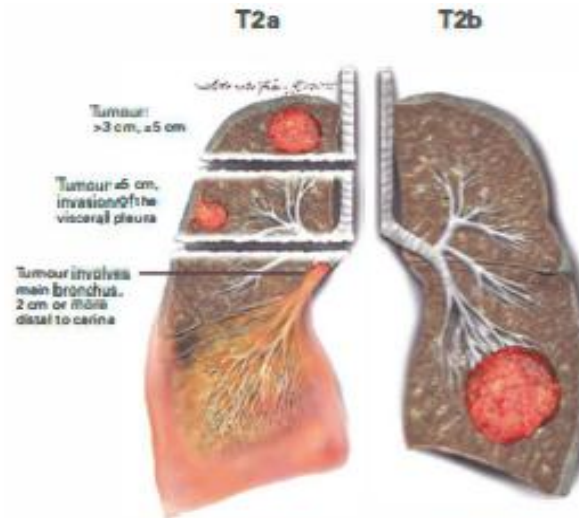
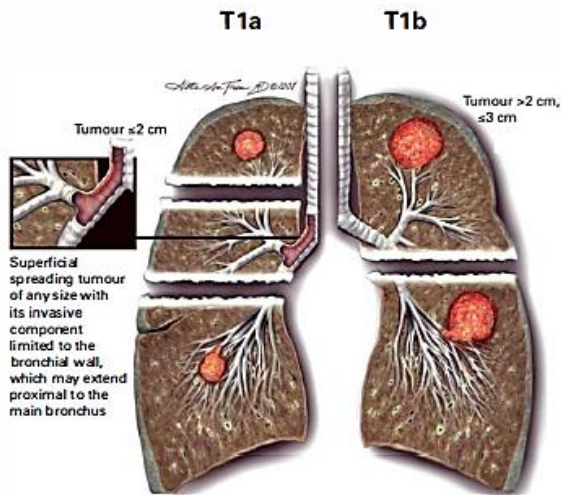
Squamous carcinoma, large cell carcinoma, adenocarcinoma.

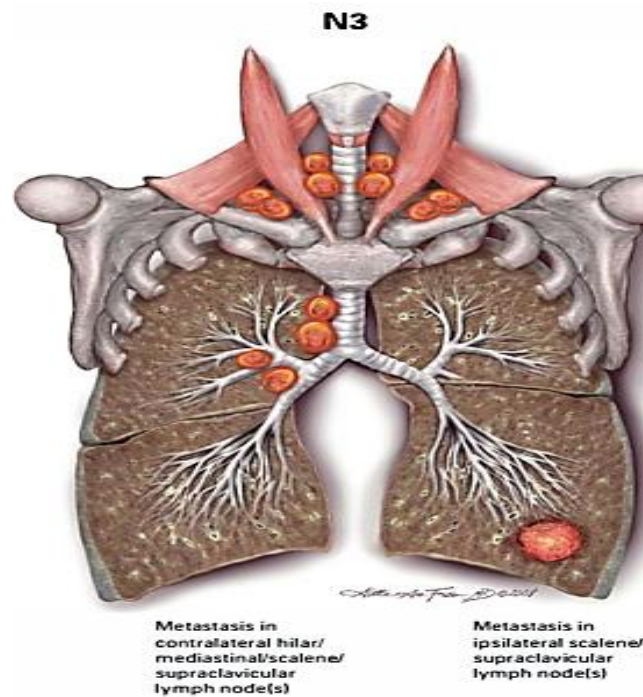
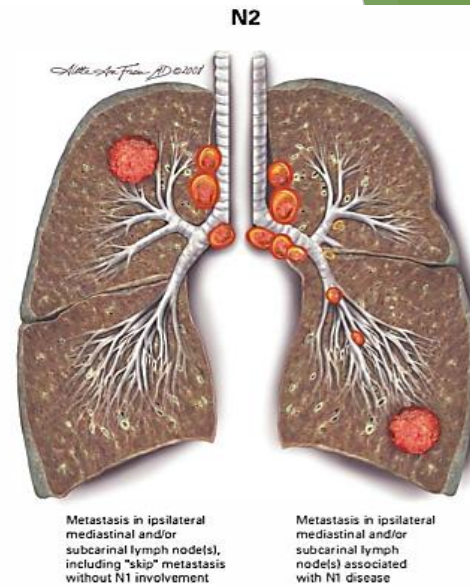
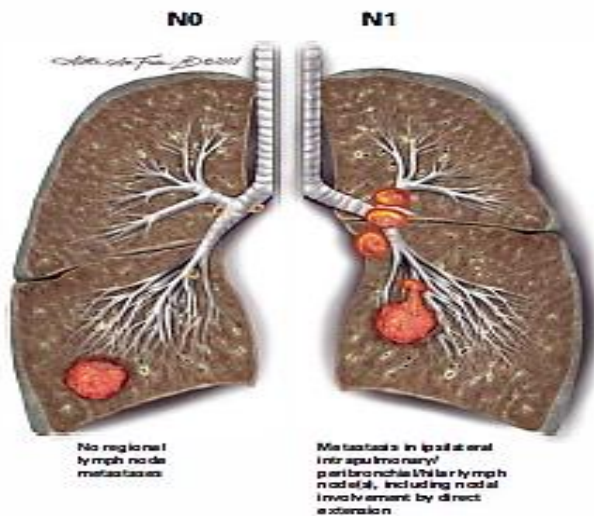
Small cell lung cancer (SCLC) (15%)

Also called 'oat cell'.

Spread

- Direct to pleura, recurrent laryngeal nerve, pericardium, oesophagus, brachial plexus.
- Lymphatic to mediastinal, supraclavicular and cervical nodes.
- Haematogenous to liver, bone, brain, adrenals.
- Transcoelomic pleural seedlings and effusion.





Investigations

Diagnostic

- Chest X-ray – PA and lateral (lung opacity, hilar lymphadenopathy).
- CT-guided lung biopsy.
- Sputum cytology.
- Bronchoscopy and cytology of brushings or lavage fluid.

Assess operability

- Helical CT scan of thorax/abdomen: involvement of adjacent structures, hepatic metastases, multiple primary lesions.
- Bone scan: metastases.
- Liver ultrasound: metastases.
- Mediastinoscopy: involvement of mediastinal nodes.
- Lung function test: likely patient tolerance of pulmonary resection.

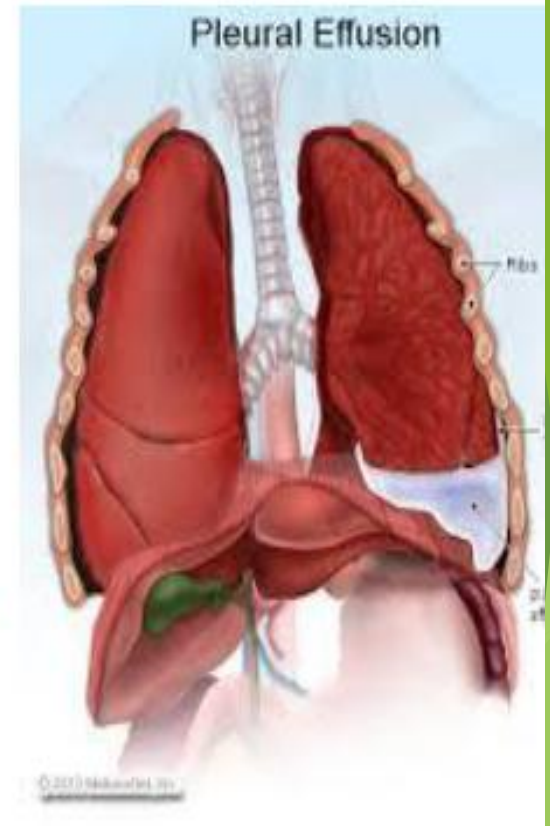
Pleural Effusion

- **Definition:**

Accumulation of fluid in pleural space.

- **Types:**

- Hydrothorax: accumulation of serous fluid in pleura
- Haemothorax: accumulation of blood in pleura
- Pyothorax: accumulation of pus in pleura
- Chylothorax: accumulation of chyle in pleura



Transudative Pleural Effusions Causes

1. Congestive Heart Failure

2. Liver Cirrhosis

3. Pulmonary Embolism

4. Nephrotic Syndrome

5. Peritoneal Dialysis

Exudative Pleural Effusions Causes

1. Neoplastic diseases

Metastatic disease

Mesothelioma

2. Infectious diseases

- ▶ Bacterial infections
- ▶ Tuberculosis
- ▶ Fungal infections
- ▶ Viral infections
- ▶ Parasitic infections

3. Pulmonary embolization

4. Esophageal perforation

5. Intra-abdominal abscesses

6. Diaphragmatic hernia

7. After abdominal surgery

How to Approach ??

Symptoms:

Asymptomatic

Dry cough

Chest pain

Dyspnea



Signs:

Inspection: limited chest movement $\pm$ bulge

Palpation: $\downarrow$ TVF $\pm$ mediastinal shift

Percussion: dullness

Auscultation: $\downarrow$ air entry $\pm$ pleural rub



No Fluid

**Fluid on the left side
(left pleural effusion)**



Biochemical Analysis

–. Diagnostic aspiration:

a. Physical examination:

- Clear yellowish to light brown fluid in hydrothorax
- Opaque red in haemothorax
- Turbid in pyothorax
- Milky white in chylothorax

b. Chemical Examination:

❖ **Light's criteria:**

To differentiate between transudate and exudate, if one criterion is positive it means exudative effusion:

1. Pleural protein > 0.5 serum protein
2. Pleural LDH > 0.6 serum LDH
3. Pleural LDH $> 2/3$ upper normal limit of serum LDH

c. Cytological examination: to exclude malignant effusion

d. Culture & sensitivity: in suspected pyothorax

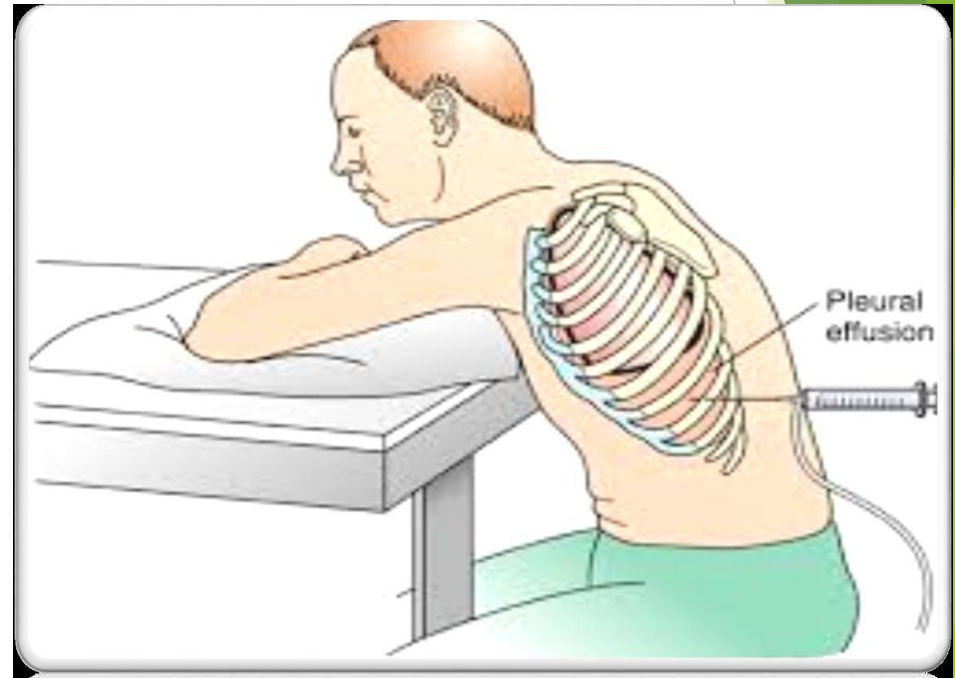
Biochemical Analysis

► If the following criteria are detected, the fluid is **exudative**.

1-PH < 7.2

2-Glucose Level < 45 gm

3-Albumin > 3.5 mg



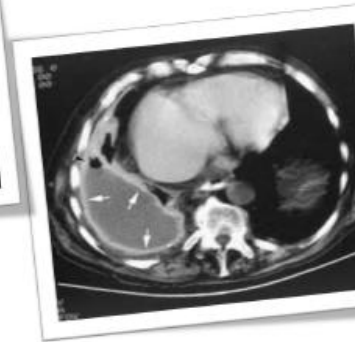
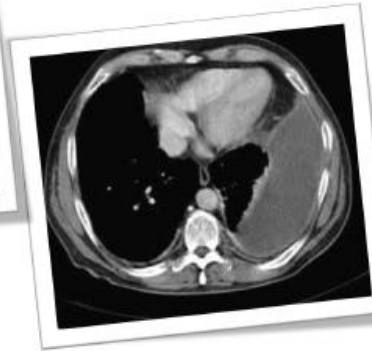
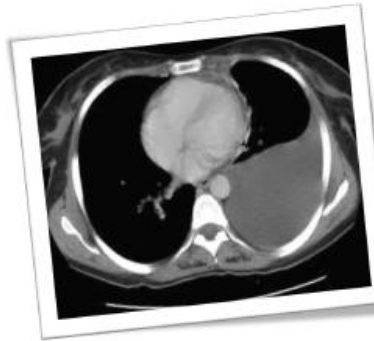
Empyema Thoracis

- **Definition:**

Empyema thoracis is a localized or free collection of purulent material in the pleural space as a result of combination of pleural dead space, culture medium of pleural fluid, and inoculation of bacteria.

- **Types of empyema:**

- *Acute empyema*: empyema in the first 14 days.
- *Subacute empyema*: empyema after 14 days till becoming chronic empyema.
- *Chronic empyema*: empyema with failure of lung expansion (entrapment of lung) despite of adequate drainage (well-functioning chest tube).



- **Etiology:**

1. Pneumonia: "post-pneumonic empyema" is the most common cause.
2. Ruptured lung abscess and Bronchiectasis.
3. Chest trauma.
4. Thoracic interventional procedures, such as thoracentesis and Esophagoscopy.
5. Extension of infection from neck, abdomen, or mediastinum.

- **Pathophysiology:**

The pathological progression in empyema can be divided into three stages:

- **Stage I: exudative phase (*para-pneumonic effusions*):**

With the same criteria of exudative pleural effusion

- **Stage II: fibropurulent phase (*complicated para-pneumonic effusions*):**

In this stage there is invasion of pleural fluid by bacteria, increased fibrin deposition, cellular debris and white blood cells with the ultimate formation of limiting fibrin membranes producing loculations.

- **Stage III: organization phase:**

Fibroblasts grow into the exudative fibrin sheet coating the viscera and parietal pleura with an inelastic membrane or pleural “peel” encasing the lung and rendering it functionless.

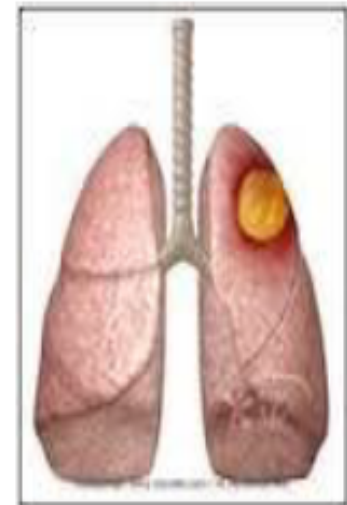
Lung Abscess

- **Definition:**

Necrosis of the pulmonary tissue and formation of cavities containing necrotic debris or fluid caused by microbial infection.

- **Clinical picture:**

- Patients often present with indolent symptoms that evolve over a period of weeks to months.
- The usual symptoms are fever, cough with sputum production (The expectorated sputum characteristically is foul smelling and bad tasting in case of anaerobic infection), night sweats, anorexia, and weight loss.
- Patients may develop hemoptysis or pleuritic chest pain.



Investigations:

1. *Chest radiography:*

- A typical chest radiographic appearance of a lung abscess is an irregularly shaped cavity with an air-fluid level inside.
- Lung abscesses as a result of aspiration most frequently occur in the posterior segment of the Rt. upper lobe or the superior segment of the Rt. lower lobe.

2. *Computed tomography:*

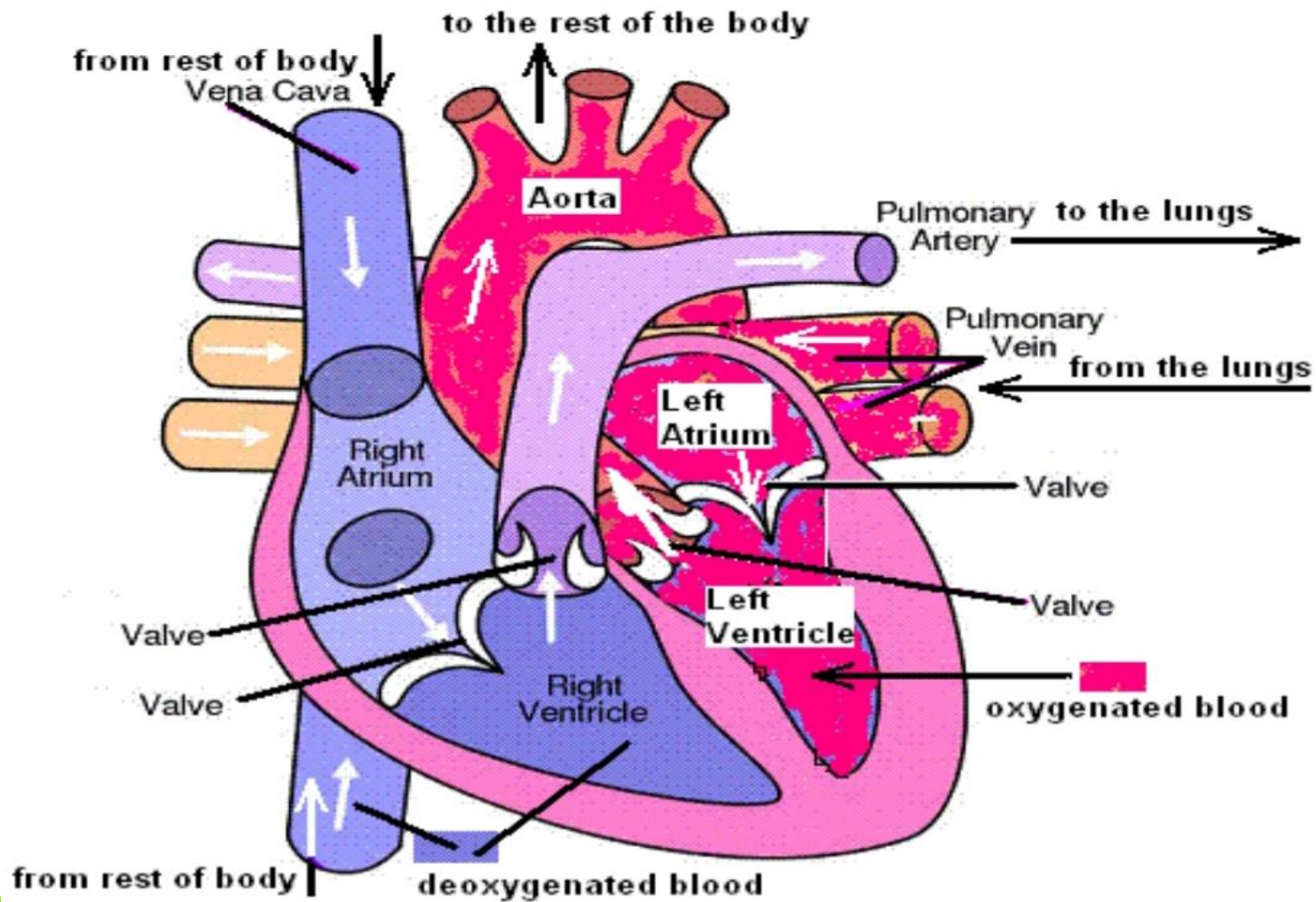
3. *Flexible fiberoptic bronchoscopy :*

4. *Laboratory Studies:*





II-Principles of Cardiac Surgery



Cardiopulmonary bypass(CPB)

CPB may be required in patients with:

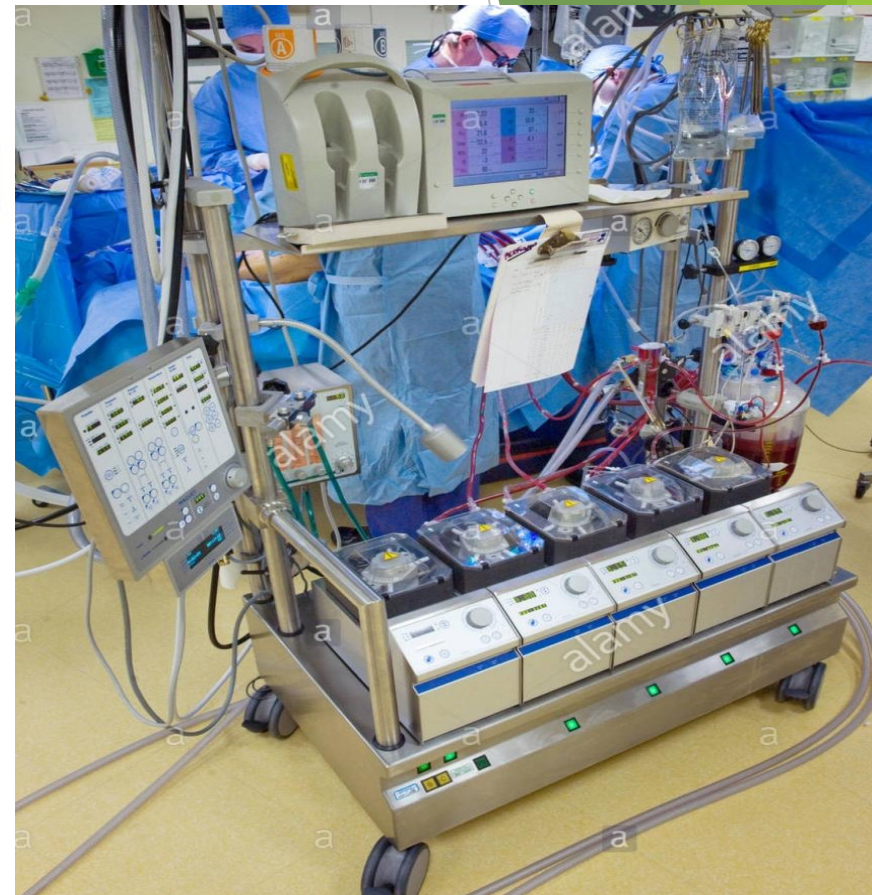
- 1-CAD,
- 2-Valvular heart disease,
- 3-Aortic dissections,
- 4- Congenital heart defects,
- 5-Aneurysms (aortic, ventricular),
- 6-Transplantation (heart, lung, trachea)

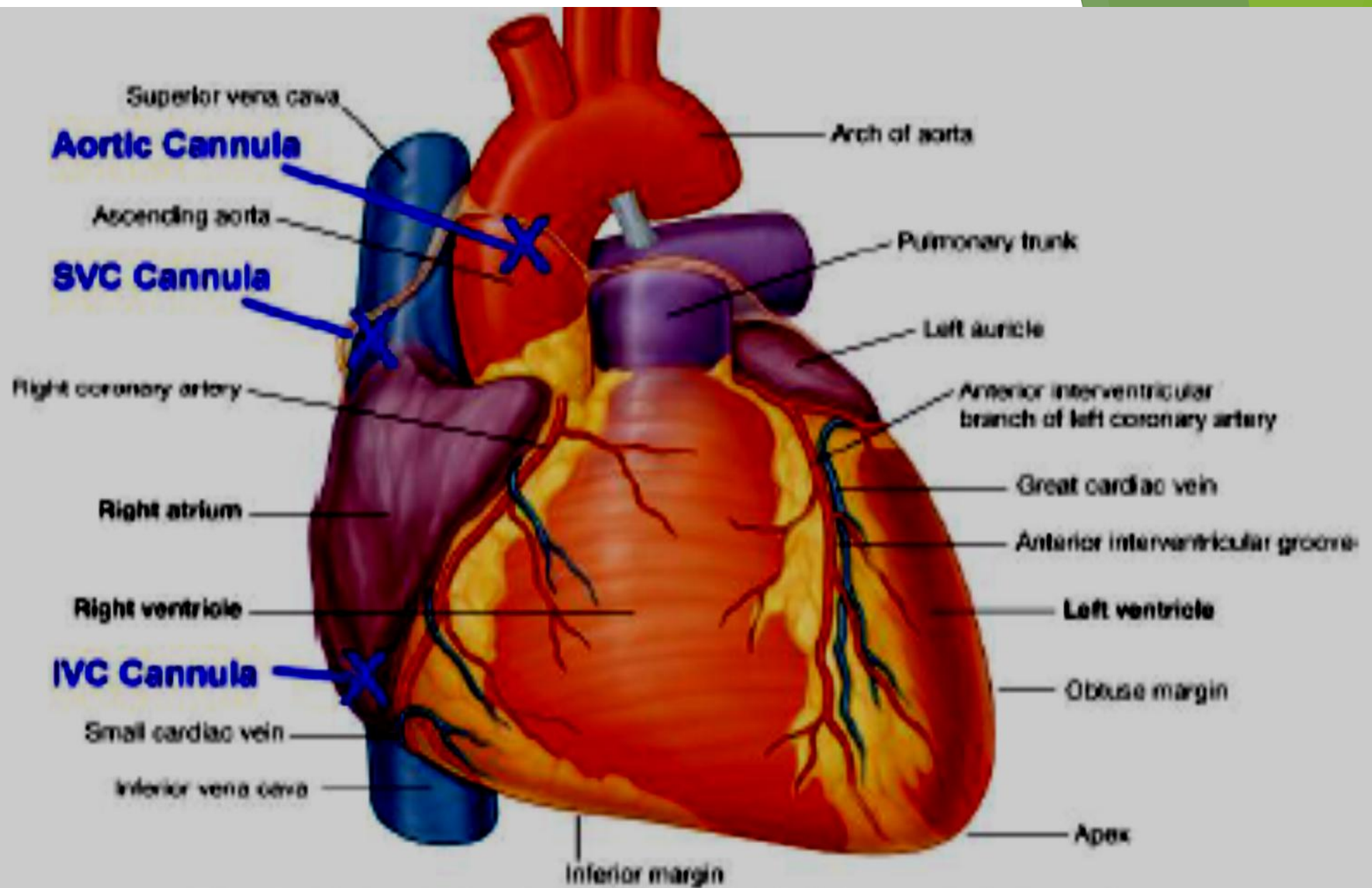
Function of CPB

- 1-CPB facilitates surgical intervention with a **motionless** and/or a **bloodless** field.
- 2-The bypass machine can isolate the heart and lungs while maintaining organ perfusion.
- 3- It both serves as **a pump** and **an oxygenator**, with pressure and oxygenation easily modulated by increasing flow rates on CPB.

Reservoir& Oxygenator

- 1- Reservoir receives blood from venous lines and the amount of veous drainage is regulated by the height of the reservoir.
- 2- Oxygenator receives venous blood from the reservoir, Blood passes through the membrane oxygenator where *CO₂/O₂ exchange* takes place, in a 40 micron filter and then back into the arterial circuit of the patient.





Cardioplegia

- ▶ Cardioplegia is the most common method of accomplishing asystole.

It can be given antegrade and/or retrograde.

- ▶ Antegrade cardioplegia is usually quicker at arresting the heart (30-60 seconds). This is done using a cardioplegia cannula that is inserted directly into **the ascending aorta** or, alternatively, directly down **the coronary ostia** following aortotomy.

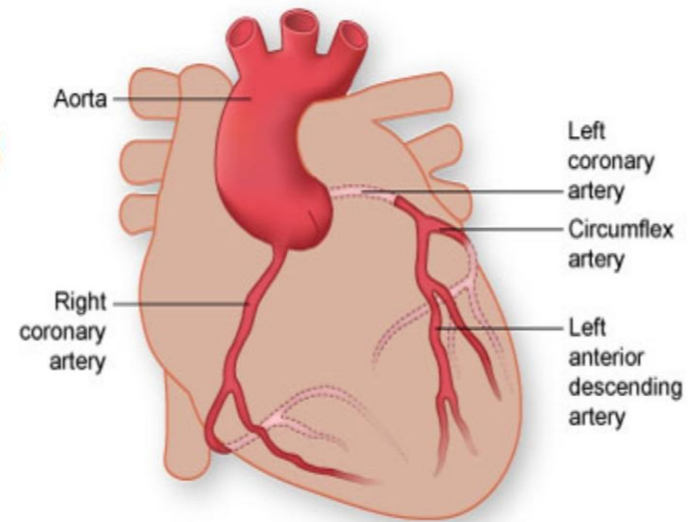
Temperature

- 1-CPB can be performed either by normothermia or cooling the patient to **30,28°** C or even low at 15-18° C FOR complete circulatory arrest.
- 2-Cardioplegia is induced by **high potassium** solution which causes electromechanical arrest of the myocardium, which decreases the myocardial O² utilization by **90%**.
- 3-An additional **10%** decrease is achieved by cooling the myocardium

Coronary arteries

► There are 3 main coronary arteries:

- 1- Right Coronary ,
- 2- Left Anterior Descending,
- 3- circumflex A.



► **Dominance** of the left or right is defined by the supplier of the posterior descending artery (PDA), which is from the RCA in 85% to 90%.

► **The left main coronary artery (LMCA)** courses off the left Sinus of Valsalva, behind the PA and anterior to the LA appendage, and bifurcates into the LAD and circumflex (or trifurcates in the setting of a ramus artery).

► **The LAD** is the interventricular artery and courses along the anterior interventricular groove towards the apex. It gives off septal perforators that dive down perpendicularly into the septum, diagonals that course over the LV anterior wall, and RV branches that course to the anterior surface of the RV.

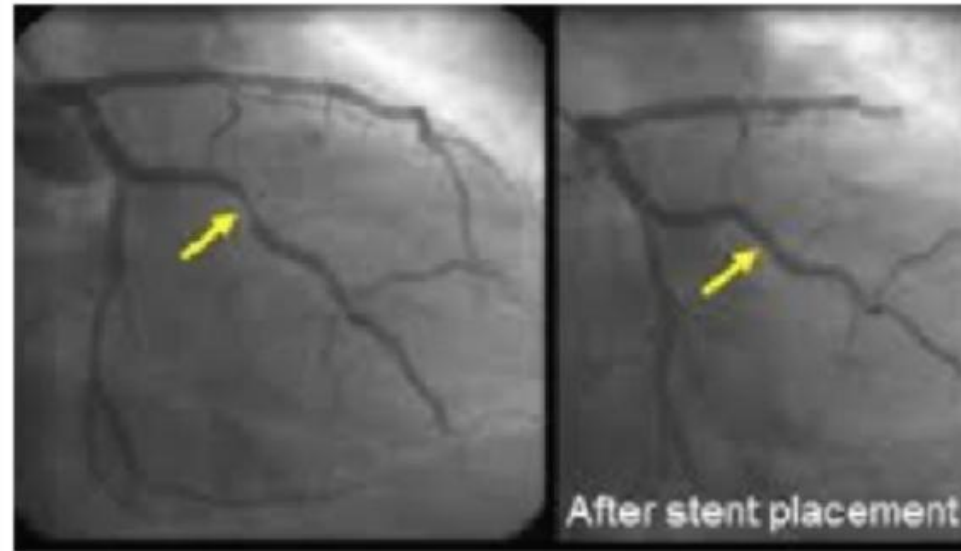
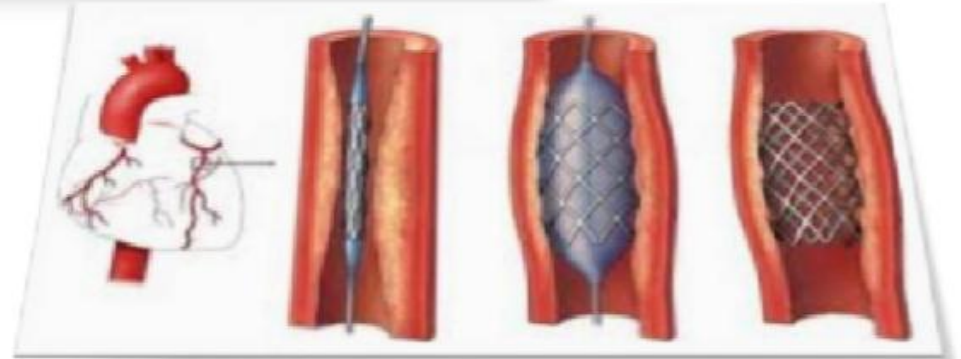
Coronary Artery Disease

• **Treatment options:**

➤ *Medical ttt*

➤ *PCI (stenting)*

➤ *Surgical ttt (CABG)*



Indications of CABG Surgery

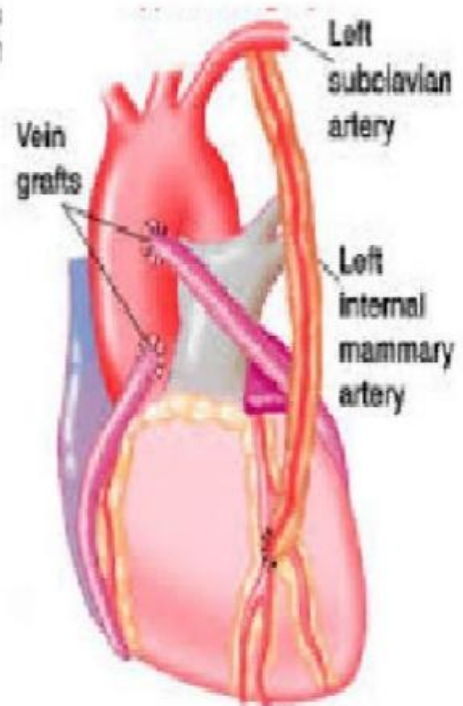
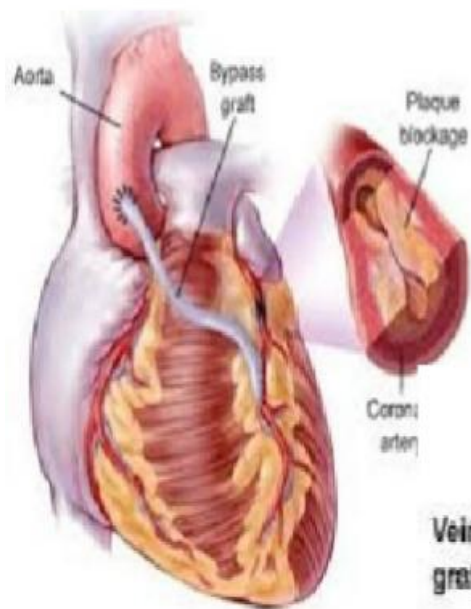
1. Triple Vessel Disease
2. Critical Left Main Stenosis $\geq 50\%$
3. Left Main Equivalent with decreased EF

Conduits for CABG

1-Internal Mammary Artery(10-year patency is 95%).

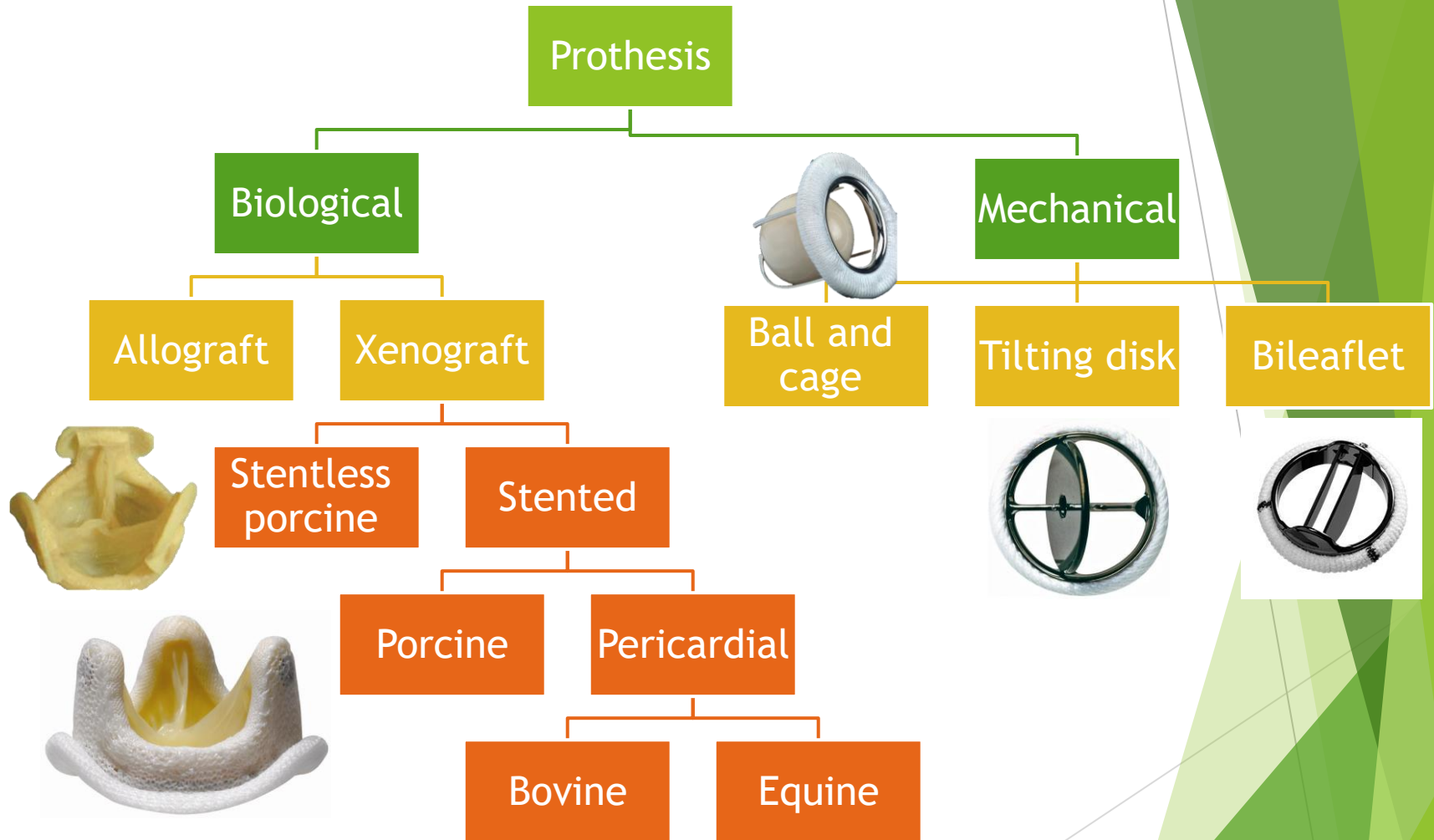
2-Long Saphenous Vein(10-year patency is 50%) due to intimal hyperplasia.

3-Radial Atery



Valvular Cardiac Surgery

Types of Heart Valve Prosthesis:



	Biological	Mechanical
Advantages	No need For anticoagulation	More Durable
Disadvantages	Less Durable	Must have anticoagulation
Indications	Contraindication for anti-coagulation Female in child-bearing period Old age > 70 years	Young Healthy patients Below 70 years Presence of other indications of anticoagulation

THANK YOU

